

Fake News & Misinformation Detection System

Final Seminar Presentation

Seminar Project Team

HCMUS

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Business Problem and Motivation

- Manual verification is slow compared to misinformation spread.
- Review teams need fast first-pass triage with evidence support.
- Goal: improve review efficiency while keeping decisions explainable.

Success targets

Business: reduce triage time. Technical: high F1 and stable latency.

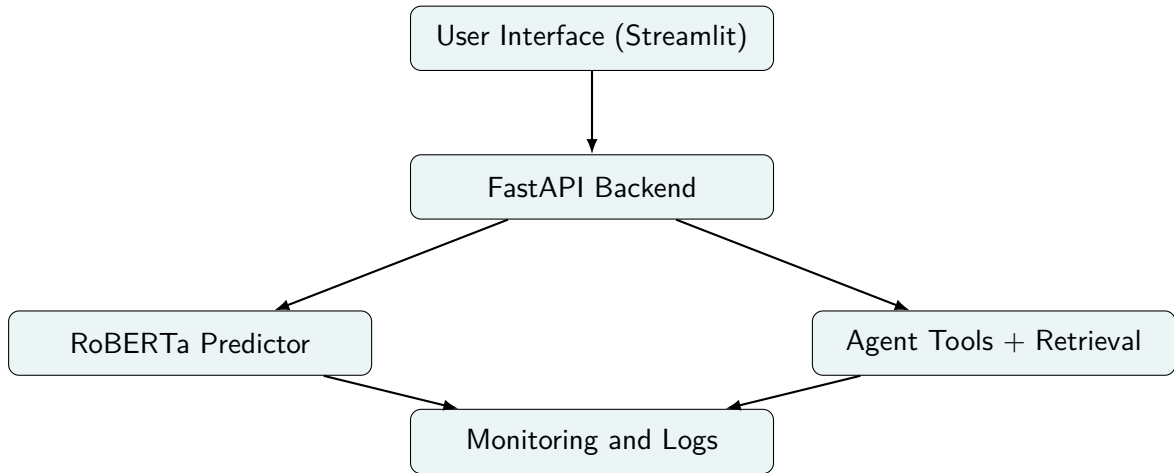
Proposed NLP Solution

- Baseline model: TF-IDF + Logistic Regression
- Main model: RoBERTa text classifier
- Agentic orchestration for evidence and decision synthesis
- FastAPI backend + Streamlit frontend

Output

Prediction, confidence, trust score, evidence summary, explanation.

System Architecture Diagram



- Dataset format: CSV with `text` and `label`.
- Labels: fake and real.
- Split strategy: 70% train, 15% validation, 15% test (stratified).
- Preprocessing: clean HTML/URL, normalize text, remove duplicates and empty rows.

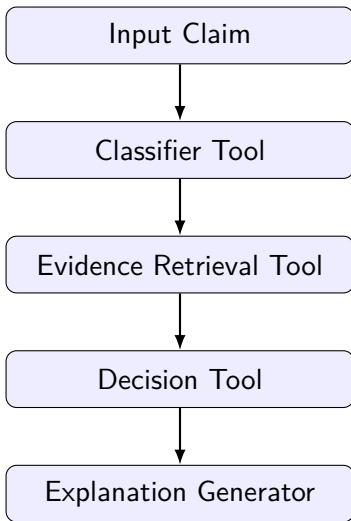
Model and Evaluation Results

Model	Accuracy	Precision	Recall	F1
TF-IDF + LR	0.9913	0.9938	0.9870	0.9904
RoBERTa	0.9991	0.9987	0.9997	0.9992

Takeaway

RoBERTa improves overall quality; baseline remains a strong benchmark.

Agentic AI Component



Final output can be REAL, FAKE, or UNCERTAIN depending on confidence and evidence conflict.

Deployment Overview

- Backend deployment: Render free tier (FastAPI service).
- Frontend deployment: Streamlit Community Cloud.
- Communication via `API_URL` environment variable.
- Runtime endpoints: `/health`, `/version`, `/predict`, `/analyze`, `/agent`.

Demo readiness

Public URL can be accessed from any machine for course presentation.

Monitoring and Continual Learning Plan

- Track confidence drift, prediction drift, uncertain rate, conflict rate, and latency.
- Collect reviewed error cases as new training candidates.
- Trigger periodic retraining when monitoring thresholds are violated.

Ethics, Privacy and Risks

- Privacy: PII masking before inference and logging.
- Ethics: false positives and false negatives can both cause harm.
- Explainability: evidence and trust score improve transparency.
- Misuse risk: over-reliance on automated output without human review.

Mitigation

Use UNCERTAIN state and human escalation for ambiguous cases.

Key Takeaways and Future Work

- Built an end-to-end NLP system aligned with production-style requirements.
- Combined strong modeling performance with deployability and agentic reasoning.
- Next steps: stronger stance modeling, broader robustness tests, and tighter MLOps automation.

Thank you

Questions and feedback